>> Engineering calculations (based on engineering mechanics) :-

→ Hellix angle (ϕ)

the major diameter from the model = 38.00 mm

the minor " " " " = 34.5 mm

∴ the average diameter of the thread $(d_{avg}) = \frac{38+34.5}{2}$

$$d_{avg} = 36.25 \text{ mm}$$

Now, we know that,

$$\tan\phi = \frac{\text{pitch}}{\pi d_{avg}}$$ (pitch = 7mm from the model).

$$\tan\phi = \frac{7}{\pi \times 36.25} = 0.0614$$

$$\phi = \tan^{-1}(0.0614) = 3.5135^\circ$$

∴ Hellix angle (ϕ) = 3.51°

→ from engineering mechanics: If the weight to be lifted by the screw jack is 'W' and coefficient of friction is 'μ' and hellix angle ϕ. Then –

The minimum ~~was~~ force required to raise the load 'W' is ⇒ $P = W \tan(\theta + \phi)$

where, $\tan\theta = \mu$ (coefficient of friction).

The force required to raise the load 'W' at the end of the tommy bar will be given by -

$$F \times \left(210 + \frac{15}{2}\right) = P \times \frac{d_{avg}}{2} \quad \text{(from the Model)}$$

$$F \times 242.5 = \frac{P \times d_{avg}}{2}$$

$$F = \frac{P \times d_{avg}}{2 \times 242.5} = \frac{P \times 36.25}{2 \times 242.5}$$

$$F = 0.074\,P \ (N)$$

$$\boxed{F = 0.07\,P} \ (N)$$

→ Efficiency per revolution:

$$\eta = \frac{W \times \text{pitch}}{P \times \pi d_{avg}}$$

$$= \frac{W \times \text{Pitch}}{W \tan(\theta+\phi)\,\pi d_{avg}}$$

$$= \frac{1}{\tan(\theta+\phi)} \times \left(\frac{\text{Pitch}}{\pi d_{avg}}\right)$$

$$\boxed{\eta = \frac{\tan\phi}{\tan(\theta+\phi)}}$$

Where: W: Load to be lifted

P: Tangential force at the mean thread radius.

→ The condition for self-locking is: $\boxed{\theta > \phi}$

Where, $\mu = \tan\phi$